10 Linux Commands:

```
ubuntu@ip-172-31-9-253:~/linux-network-lab$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 0a:ff:f4:0d:12:bf brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname enx0afff40d12bf
    inet 172.31.9.253/20 metric 100 brd 172.31.15.255 scope global dynamic ens5
       valid_lft 3121sec preferred_lft 3121sec
    inet6 fe80::8ff:f4ff:fe0d:12bf/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
ubuntu@ip-172-31-9-253:~/linux-network-lab$ ip link
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP mode DEFAULT group default qlen 1000
    link/ether 0a:ff:f4:0d:12:bf brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname enx0afff40d12bf
ubuntu@ip-172-31-9-253:~/linux-network-lab$ ip route
default via 172.31.0.1 dev ens5 proto dhcp src 172.31.9.253 metric 100
172.31.0.0/20 dev ens5 proto kernel scope link src 172.31.9.253 metric 100
172.31.0.1 dev ens5 proto dhcp scope link src 172.31.9.253 metric 100
172.31.0.2 dev ens5 proto dhcp scope link src 172.31.9.253 metric 100
ubuntu@ip-172-31-9-253:~/linux-network-lab$ hostname -I
172.31.9.253
ubuntu@ip-172-31-9-253:~/linux-network-lab$ getent hosts example.com
2606:4700:10::ac42:93f3 example.com
2606:4700:10::6814:179a example.com
ubuntu@ip-172-31-9-253:~/linux-network-lab$ cat /etc/resolv.conf
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-resolved(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
```

```
nameserver 127.0.0.53
options edns0 trust-ad
search ap-south-1.compute.internal
ubuntu@ip-172-31-9-253:~/linux-network-lab$ ip neigh
172.31.0.2 dev ens5 lladdr 0a:ff:e2:7d:67:33 STALE
172.31.0.1 dev ens5 lladdr 0a:ff:e2:7d:67:33 REACHABLE
ubuntu@ip-172-31-9-253:~/linux-network-lab$ ss -tuln
Netid            State             Recv-Q            Send-Q                        Local Address:Port                        Peer Address:Port
udp              UNCONN            0                 0                               127.0.0.54:53                             0.0.0.0:*
udp              UNCONN            0                 0                            127.0.0.53%lo:53                             0.0.0.0:*
udp              UNCONN            0                 0                         172.31.9.253%ens5:68                             0.0.0.0:*
udp              UNCONN            0                 0                               127.0.0.1:323                              0.0.0.0:*
udp              UNCONN            0                 0                                   [::1]:323                                 [::]:*
tcp              LISTEN            0                 4096                            127.0.0.54:53                             0.0.0.0:*
tcp              LISTEN            0                 4096                         127.0.0.53%lo:53                             0.0.0.0:*
tcp              LISTEN            0                 4096                               0.0.0.0:22                             0.0.0.0:*
tcp              LISTEN            0                 4096                                  [::]:22                                [::]:*
ubuntu@ip-172-31-9-253:~/linux-network-lab$ ss -tun
Netid             State              Recv-Q             Send-Q                      Local Address:Port                      Peer Address:Port
tcp               ESTAB              0                  0                         172.31.9.253:22                     13.233.177.5:44690
```

1. `ip addr` – Displays network interfaces and their assigned IP addresses.
2. `ip link` – Shows network interfaces and their operational states.
3. `ip route` – Displays the routing table and default gateway.
4. `hostname -I` – Prints the IP addresses assigned to the system.
5. `getent hosts example.com` – Resolves a hostname into its associated IP address using system name services.
6. `cat /etc/resolv.conf` – Displays the DNS resolver configuration.
7. `ip neigh` – Shows the neighbor cache containing discovered local network devices.

8. `ss -tuln` – Lists listening TCP and UDP sockets with numeric addresses and ports.
9. `ss -tun` – Displays TCP and UDP socket information using numeric addresses and ports.
10. `ip route` and `hostname -I` – Help compare Linux network addresses and routing with the Windows host when using WSL.